

# Shiv Nadar University Chennai

## CS3807 – Deep Learning Laboratory

### Experiment 7

#### End-to-End Study of Autoencoders, Convolutional Autoencoders, Denoising Autoencoders and Variational Autoencoders

|                     |                                               |             |
|---------------------|-----------------------------------------------|-------------|
| Degree & Branch     | B.Tech Artificial Intelligence & Data Science | Semester V  |
| Subject Code & Name | CS3807 – Deep Learning Laboratory             | AY: 2026–27 |

## 1. Objective

The objective of this experiment was to study autoencoders and their variants for image reconstruction, denoising and generative modelling. Four models were implemented using MNIST: a Fully Connected Autoencoder (FC-AE), a Convolutional Autoencoder (CAE), a Denoising CAE and a Variational Autoencoder (VAE). Their behaviour was studied using reconstruction metrics, visual comparisons and latent-space analysis.

## 2. Learning Outcomes

After completing this experiment, the following concepts were studied:

- encoder–latent–decoder based image reconstruction;
- fully connected and convolutional autoencoders;
- image denoising using controlled noise;
- MSE, MAE and SSIM for reconstruction evaluation;
- probabilistic latent representations in a VAE;
- the reparameterization trick and KL-divergence loss;
- VAE image generation and latent-space interpolation; and
- the difference between deterministic and probabilistic latent spaces.

## 3. Dataset and Experimental Setup

The MNIST handwritten digit dataset was used throughout the experiment. The images are grayscale and have a resolution of  $28 \times 28$  pixels. A subset of 10,000 images was used for training and 2,000 images for testing.

Pixel values were converted from  $[0, 255]$  to floating-point values in  $[0, 1]$ . A channel dimension was added for convolutional models, giving an input shape of  $28 \times 28 \times 1$ . For the fully connected model, the images were flattened into 784-dimensional vectors.

All models were trained for 20 epochs with a batch size of 128 wherever applicable. The Adam optimizer with learning rate  $10^{-3}$  was used. A fixed random seed of 42 was used to improve reproducibility.

## 4. Overall Experimental Pipeline

The experiment followed the common pipeline:

MNIST  $\rightarrow$  Preprocessing  $\rightarrow$  Encoder  $\rightarrow$  Latent Representation  $\rightarrow$  Decoder  $\rightarrow$  Reconstruction  $\rightarrow$  Evaluation.

The FC-AE and CAE were first used for reconstruction. The CAE was then trained with noisy inputs for denoising. Finally, the VAE was used to learn a probabilistic two-dimensional latent space and generate new images.

## 5. Fully Connected Autoencoder

The FC-AE received a flattened 784-dimensional image and compressed it to a 16-dimensional latent representation. The architecture was

$$784 \rightarrow 128 \rightarrow 32 \rightarrow 16 \rightarrow 32 \rightarrow 128 \rightarrow 784.$$

ReLU was used in the hidden layers and sigmoid was used at the output. The model was trained using binary cross-entropy reconstruction loss.

### Plot 1: Original vs. Reconstructed Images

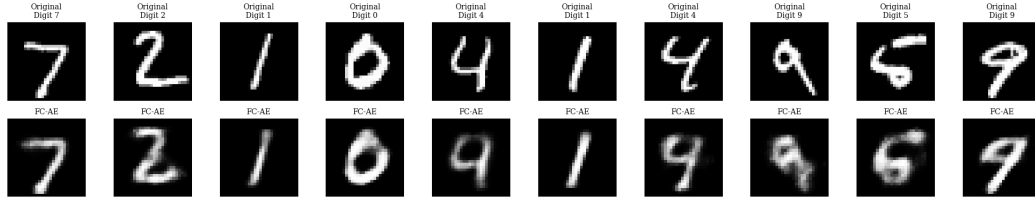


Figure 1: Original MNIST images and FC-AE reconstructions.

**Inference:** The model retains the overall form and identity of most numbers, but some of the strokes look smoother than in the actual images. This is due to the reason that the whole image has been compressed into just 16 latent variables.

### Plot 2: Training and Validation Loss

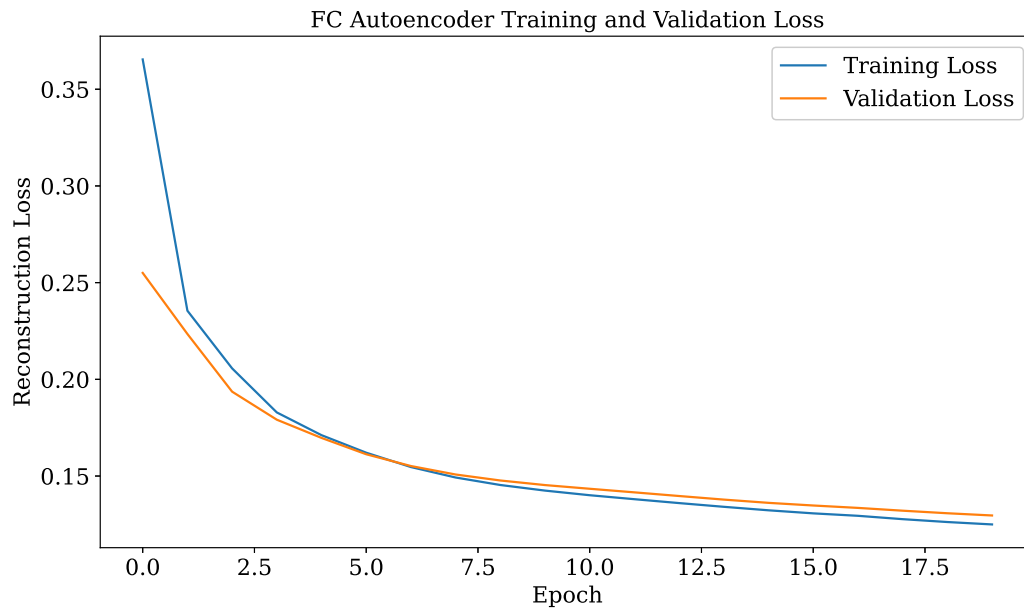


Figure 2: FC-AE training and validation reconstruction loss.

**Inference:** Both losses are decreasing steadily during the training process. In the last epoch, the training loss achieved approximately 0.1250 and the validation loss approximately 0.1296. The distances between them are similar, which means that there are no signs of overfitting.

## Reconstruction Metrics

Table 1: FC-AE reconstruction performance on the test set.

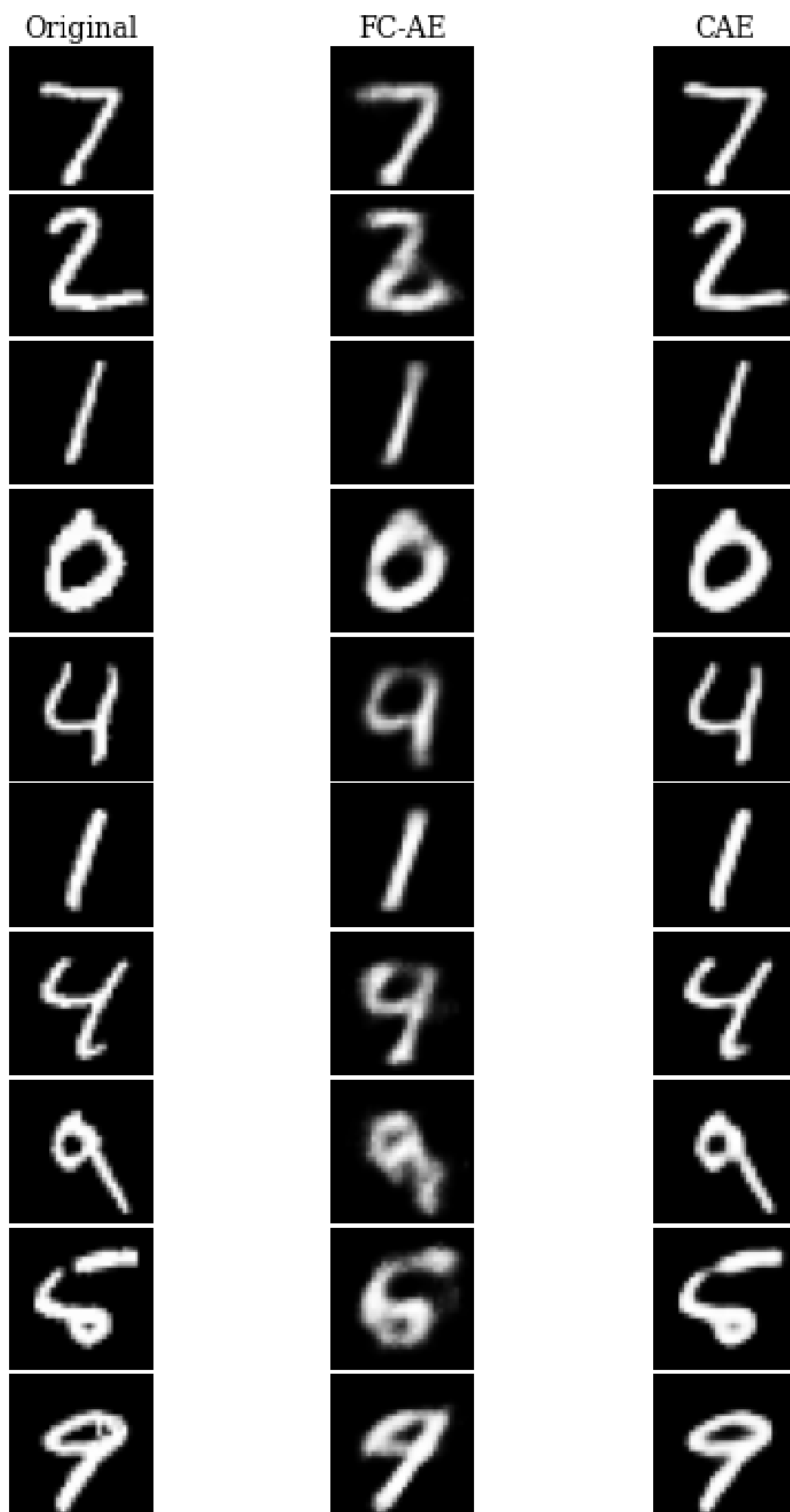
| Metric            | Value    |
|-------------------|----------|
| MSE               | 0.022520 |
| MAE               | 0.059343 |
| Mean SSIM         | 0.730456 |
| Parameters        | 211,040  |
| Training Time (s) | 11.87    |

The relatively larger reconstruction error compared with the convolutional model reflects the limitation of flattening the image: the dense network does not explicitly preserve neighbouring pixel relationships.

## 6. Convolutional Autoencoder

The CAE retained the spatial structure of the images. Its encoder used convolution and pooling layers, while the decoder used upsampling and convolution layers. The model was trained for 20 epochs.

Plot 3: FC-AE vs. CAE Reconstruction



**Inference:** CAE reconstruction has more accuracy in preserving strokes and boundaries than FC-AE. This is because convolutional filters work on local neighborhoods, thus maintaining spatial information. This makes the convolutional network more appropriate for image reconstruction.

## CAE Metrics

Table 2: CAE reconstruction performance.

| Metric            | Value    |
|-------------------|----------|
| MSE               | 0.002682 |
| MAE               | 0.015420 |
| Mean SSIM         | 0.972842 |
| Parameters        | 74,497   |
| Training Time (s) | 19.80    |

The CAE uses substantially fewer parameters than the FC-AE while producing a much lower reconstruction error and higher SSIM. The training time is slightly higher because convolutional operations are more computationally involved.

## 7. Denoising Convolutional Autoencoder

The denoising CAE was trained using corrupted images as inputs and clean MNIST images as targets. Gaussian noise was generated as

$$\tilde{x} = x + n, \quad n \sim \mathcal{N}(0, \sigma^2),$$

and the resulting pixels were clipped to  $[0, 1]$ .

The model was trained using  $\sigma = 0.2$ . Gaussian noise levels of 0.1, 0.2 and 0.3 were then evaluated. Salt-and-pepper noise with probabilities 0.05, 0.10 and 0.20 was also implemented as a second corruption type.

Plot 4: Clean, Noisy and Denoised Images

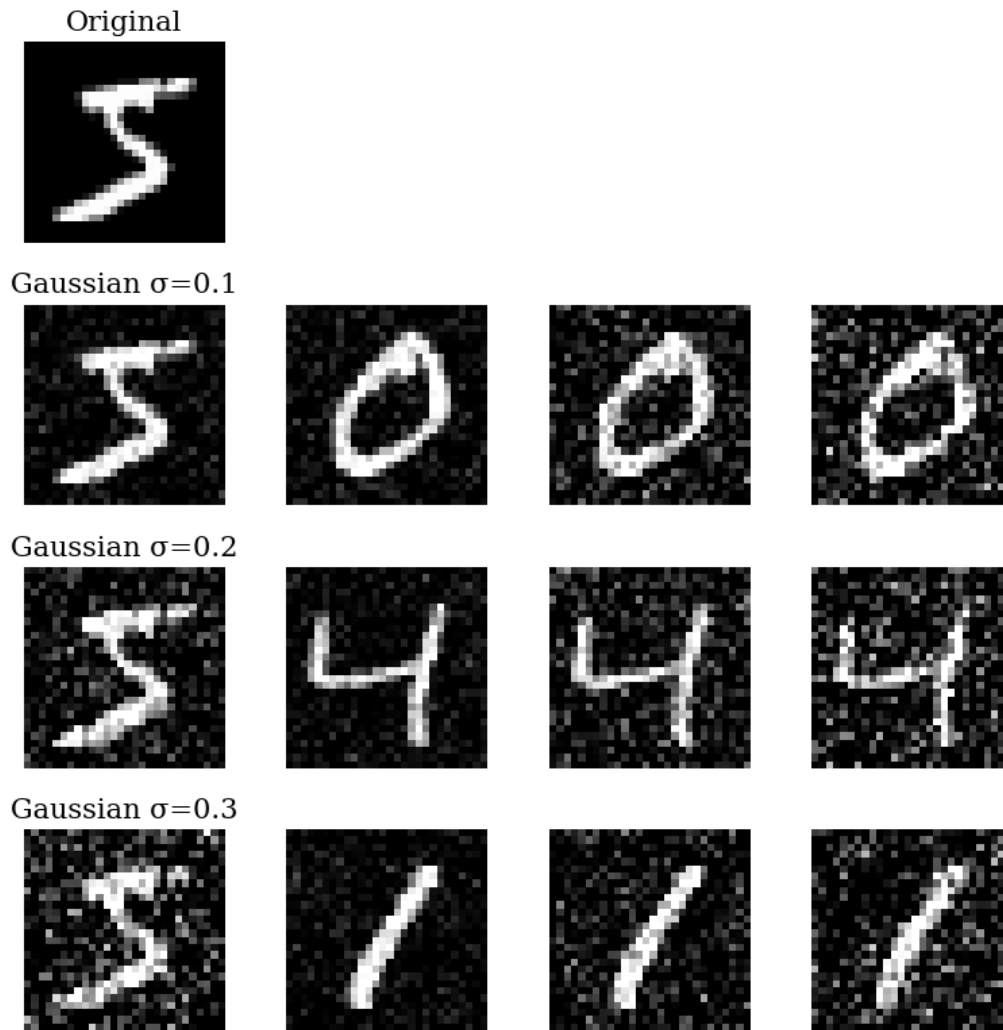


Figure 4: Clean, noisy and denoised MNIST images.

**Inference:** The denoising CAE is able to remove most of the noise from the image, while keeping the basic shape of the digit intact. The reconstructed lines are somewhat smoother but still look like the digit. This means that the network has learned structural features and has not copied the input.

## Plot 5: Noise Level vs. Reconstruction Metrics

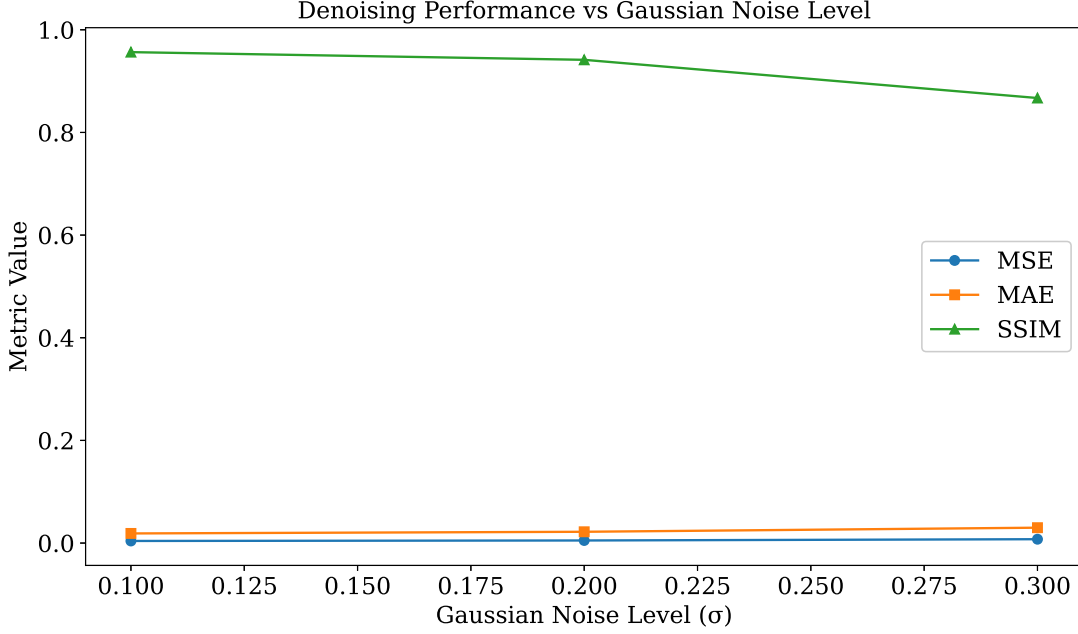


Figure 5: Effect of Gaussian noise level on reconstruction metrics.

**Inference:** The measured results show a clear deterioration as the Gaussian noise level increases:

| $\sigma$ | MSE      | MAE      | SSIM     |
|----------|----------|----------|----------|
| 0.1      | 0.004103 | 0.018759 | 0.956461 |
| 0.2      | 0.005051 | 0.021984 | 0.941483 |
| 0.3      | 0.007523 | 0.029951 | 0.867058 |

MSE and MAE increase with stronger noise, while SSIM decreases. This is expected because stronger corruption removes more information from the input, making accurate recovery harder.

## Denoising CAE Metrics

Table 3: Denoising CAE performance for  $\sigma = 0.2$ .

| Metric            | Value    |
|-------------------|----------|
| MSE               | 0.005072 |
| MAE               | 0.022050 |
| Mean SSIM         | 0.941061 |
| Parameters        | 74,497   |
| Training Time (s) | 16.96    |

## 8. Variational Autoencoder

The VAE learns a probability distribution in the latent space instead of a single deterministic latent vector. The encoder produces a mean  $\mu$  and log-variance  $\log \sigma^2$ . A latent sample is obtained using the reparameterization trick:

$$z = \mu + \sigma \odot \epsilon, \quad \epsilon \sim \mathcal{N}(0, I).$$

The VAE used a two-dimensional latent space. Its objective was

$$L_{\text{VAE}} = L_{\text{rec}} + L_{\text{KL}},$$

where the reconstruction loss encourages accurate image recovery and the KL-divergence term regularizes the latent distribution towards the standard normal prior.

**Plot 6: VAE Latent Space**

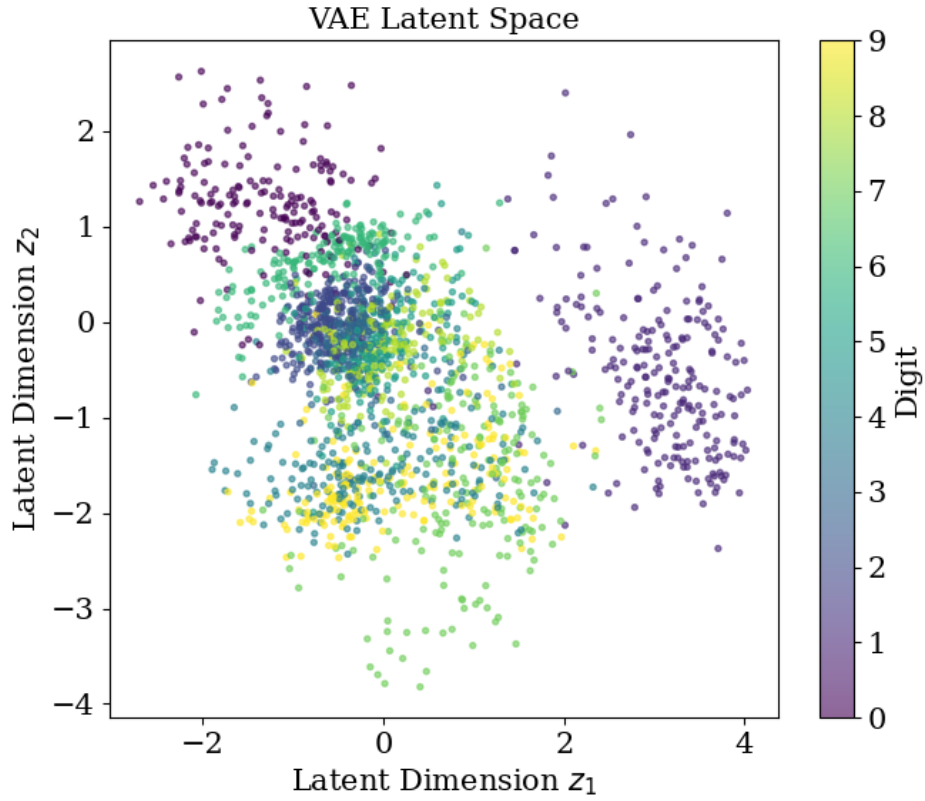


Figure 6: Two-dimensional latent space learned by the VAE.

**Inference:** The images of the same digits usually fall into close locations in the latent space. Moreover, there is a bit of intersection among digits, which is quite logical as many handwritten digits have common visual characteristics. The consecutive distribution of points allows intermediate samples generation.

### Plot 7: Generated Samples

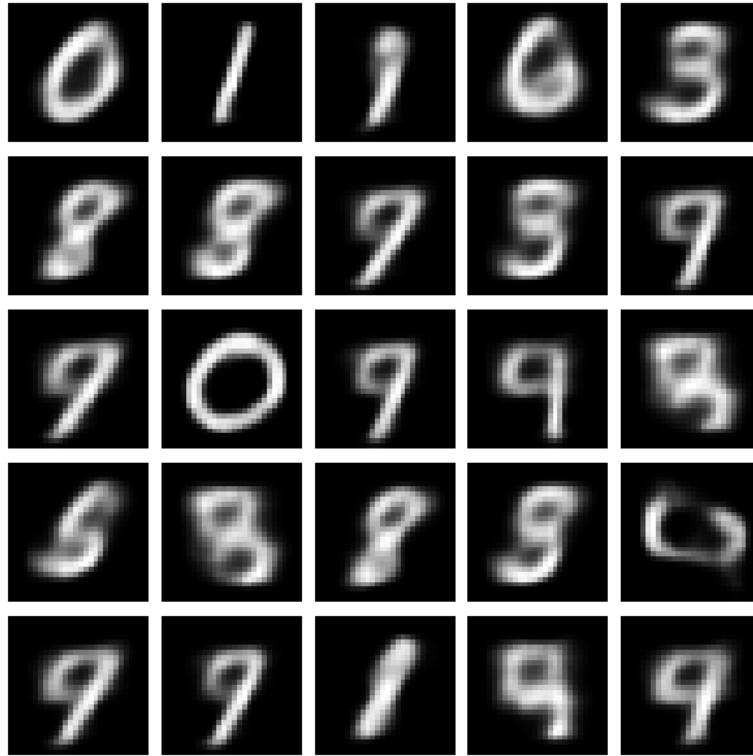


Figure 7: Twenty-five images generated by the VAE.

**Inference:** The decoder generates images that look like handwritten digits, and they are generated by sampling latent vectors from the standard normal distribution. There is variability in shapes and handwriting styles in these samples. Sometimes the samples can be not so clear, but that is expected since the images are generated from a probabilistic distribution.

### Plot 8: Latent-Space Interpolation

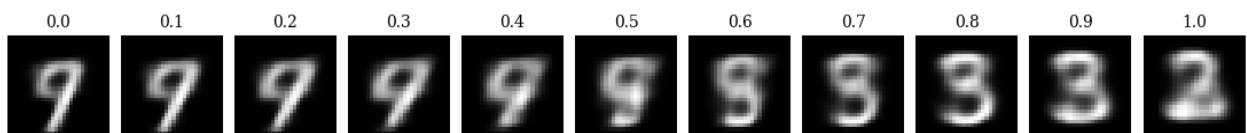


Figure 8: Decoded images obtained by interpolating between two latent points.

**Inference:** The decoded images vary in a gradual manner as the latent vector transitions from one point to another. This is an indication that the VAE model has acquired a latent space that varies continuously instead of a discrete latent space.

Plot 9: VAE Reconstruction Loss

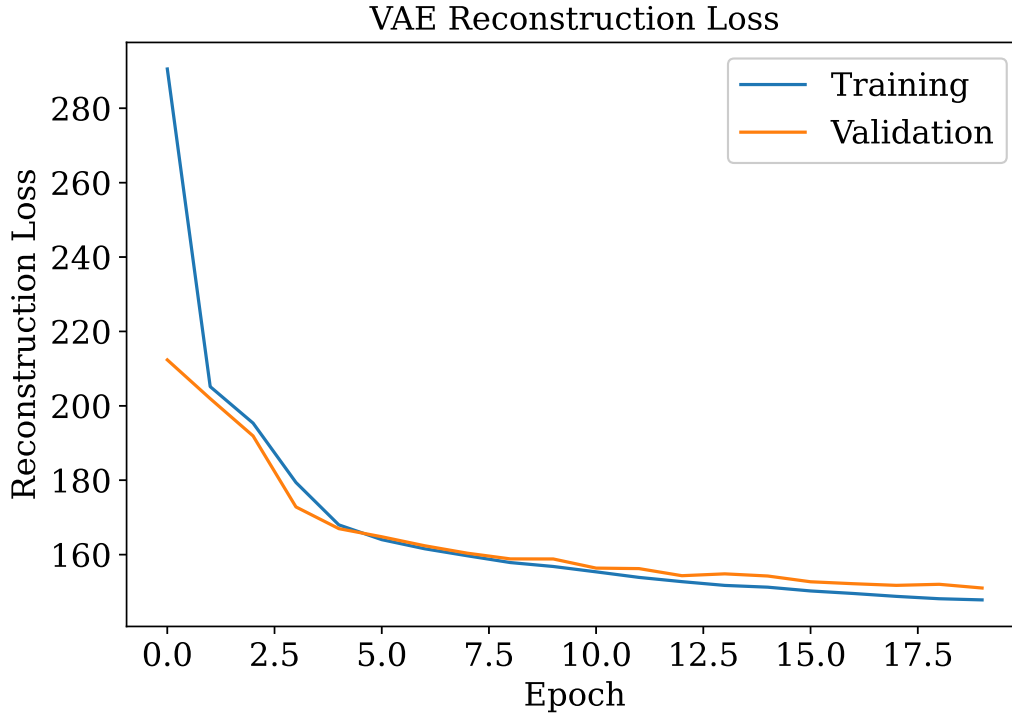


Figure 9: VAE training and validation reconstruction loss.

**Inference:** Loss due to reconstruction falls as the process of training proceeds, and validation follows the same pattern. The final value of training reconstruction loss is 147.8220, while the validation reconstruction loss is 151.0163. The small difference shows that the learned representation is also applicable for test images.

## VAE Loss Summary

Table 4: Final VAE loss components.

| Loss           | Training   | Validation |
|----------------|------------|------------|
| Reconstruction | 147.822037 | 151.016327 |
| KL Divergence  | 5.672087   | 5.895393   |
| Total VAE Loss | 153.494064 | 156.911743 |

The KL term is much smaller than the reconstruction term in this run, but it is essential because it regularizes the latent distribution and allows sampling from the learned space.

## 9. Reconstruction Error Distribution

The reconstruction error was calculated separately for each test image using MSE.

Plot 10: Reconstruction Error Distribution

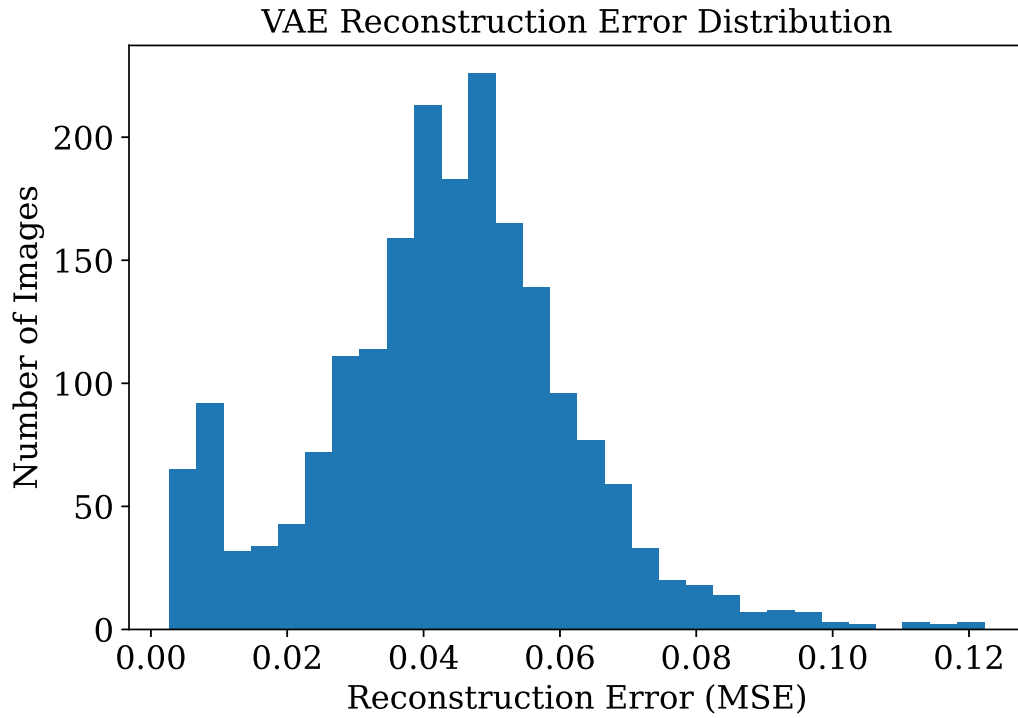


Figure 10: Distribution of VAE reconstruction errors on the test set.

**Inference:** The average value of reconstruction error is 0.043338, with a median of 0.043813 and standard deviation of 0.018815. Reconstruction errors vary between 0.002769 and 0.122247. It is notable that the majority of samples fall into the center part, whereas few have higher values of reconstruction errors.

### High-Error Samples

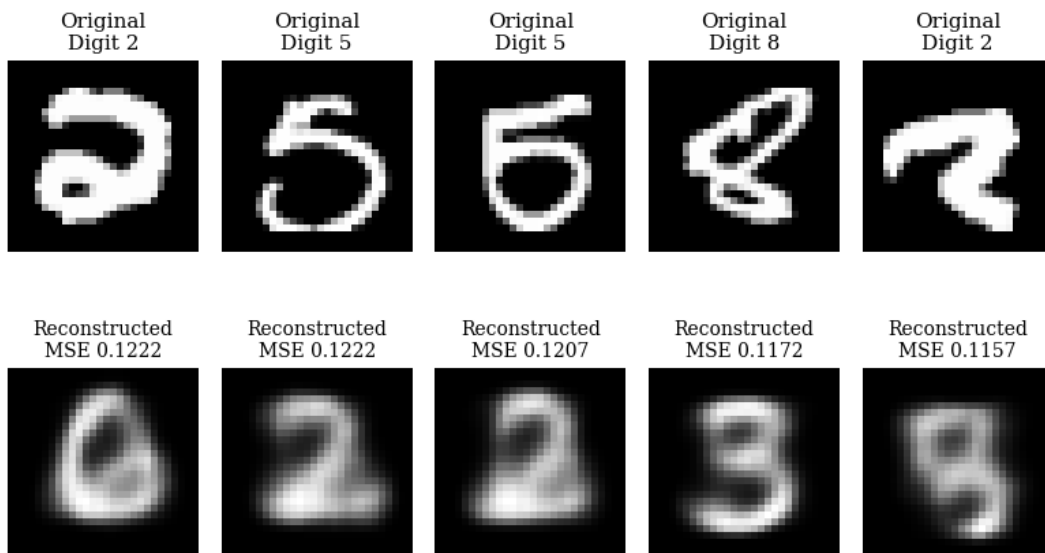


Figure 11: Five test images with the highest VAE reconstruction errors.

The five highest-error samples were:

| Rank | Test Index | Digit | MSE      |
|------|------------|-------|----------|
| 1    | 1352       | 2     | 0.122247 |
| 2    | 654        | 5     | 0.122180 |
| 3    | 1847       | 5     | 0.120664 |
| 4    | 1325       | 8     | 0.117212 |
| 5    | 1790       | 2     | 0.115693 |

**Inference:** These examples are from rather challenging hand-written images. A large amount of errors on the pixel level does not always imply that the reconstructed number cannot be recognized at all. Small stroke differences can greatly affect the MSE score although the structure of the number itself remains readable.

## 10. Consolidated Results

Table 5: Comparison of all implemented models.

| Model         | MSE      | MAE      | SSIM     | Parameters | Time (s) |
|---------------|----------|----------|----------|------------|----------|
| FC-AE         | 0.022520 | 0.059343 | 0.730456 | 211,040    | 11.87    |
| Conv-AE       | 0.002682 | 0.015420 | 0.972842 | 74,497     | 19.80    |
| Denoising CAE | 0.005072 | 0.022050 | 0.941061 | 74,497     | 16.96    |
| VAE           | 0.043338 | 0.101183 | 0.491800 | 522,885    | 25.23    |

The values show the different goals of the models. The basic CAE achieved strong reconstruction quality while using fewer parameters than the FC-AE. The denoising CAE also retained good structural similarity despite receiving corrupted inputs. The VAE has a different objective: in addition to reconstruction, it regularizes the latent distribution so that new samples can be generated.

## 11. Required Inferences

- **Latent representation:** A smaller latent space forces stronger compression and can remove fine details. The 16-dimensional FC-AE latent space was sufficient to preserve the main digit structure but not every fine feature.
- **FC-AE vs. CAE:** The CAE preserved image structure more effectively because convolution operates on local spatial neighbourhoods. In the measured results, the CAE had MSE 0.002682 and SSIM 0.972842, compared with MSE 0.022520 and SSIM 0.730456 for the FC-AE.
- **Gaussian noise:** Increasing  $\sigma$  from 0.1 to 0.3 increased MSE from 0.004103 to 0.007523 and reduced SSIM from 0.956461 to 0.867058. Therefore stronger corruption makes reconstruction harder.
- **Denoising ability:** The denoising CAE recovered the major digit structure from noisy inputs. The reconstructed images were smoother, but the important strokes were retained.
- **VAE latent distribution:** The VAE does not assign one fixed latent vector to each image. Instead, it learns a mean and variance and samples from the resulting distribution. This makes the representation suitable for generation.
- **Two-dimensional latent space:** The two-dimensional representation forms meaningful regions for different digit types, while some overlap occurs between visually similar digits.
- **Generated samples:** Sampling from the standard normal latent distribution produced recognizable digit-like images with variation in appearance. Some samples were less clear, reflecting the probabilistic nature of the model.
- **Interpolation:** The interpolation plot shows gradual visual changes between two latent points. This supports the idea that the VAE has learned a continuous latent space.
- **Reconstruction error and visual quality:** A higher MSE generally indicates greater pixel-level difference, but it does not always mean that the complete digit is visually unrecognizable. Small stroke shifts can produce large pixel errors.
- **Deterministic vs. probabilistic representation:** The conventional autoencoders learn deterministic latent representations. The VAE learns a probability distribution, which makes sampling, generation and interpolation possible.

## 12. Source Code

The experiment was implemented in Python using TensorFlow/Keras, NumPy, Matplotlib, scikit-image and the MNIST dataset.

The complete executable implementation, including the convolutional autoencoder, denoising model, VAE, evaluation code and EPS plot generation, is contained in the following GitHub Link .

<https://github.com/Siddharth-G06/Deep-Learning-Lab/tree/main/Lab%207%20-%20End-to-End%20Study%20of%20Autoencoders%2C%20Convolutional%20Autoencoders%2C%20Denoising%20Autoencoders%20and%20Variational%20Autoencoders>

## 13. Discussion Questions

### 1. What is an autoencoder?

An autoencoder is a neural network that is trained to compress an input into a compact form and then decode this compressed data into its original form.

### 2. What is the purpose of the encoder?

The encoder translates the input image into a smaller representation with all of its important data.

### 3. What is the purpose of the decoder?

The decoder translates the latent representation back into an image that resembles the input.

### 4. What is a latent representation?

This is the compact internal representation created by the encoder.

### 5. Why is the latent representation smaller than the input?

Smaller size means that the model will focus on the important information in the input.

### 6. What happens when the latent dimension is too small?

Too much information is lost, which makes the reconstruction worse.

### 7. What happens when the latent dimension is increased?

More information can be preserved, but the compression gets weaker.

### 8. Why can a convolutional autoencoder preserve spatial information better?

Convolutional filters work with neighbourhood pixels at once and learn local spatial features such as edges and strokes.

### 9. Differentiate an autoencoder and denoising autoencoder.

The regular autoencoder works with the same input data, while the denoising autoencoder is given the corrupted image and needs to reconstruct the clean one.

### 10. Why is the clean image used as the target in denoising?

It helps the model understand what the image looks like without the artificial noise.

### 11. What happens when the noise level is increased?

It becomes more challenging to reconstruct the image. In our experiment, MSE and MAE values increase while SSIM decreases when the Gaussian noise is increased.

### 12. Why are MSE, MAE and SSIM useful for image reconstruction?

MSE calculates the square error of pixels, MAE calculates the absolute value of error of pixels, and SSIM calculates the structural similarity of images. All together, they give a more complete evaluation of the quality of reconstruction.

### 13. What is Variational Autoencoder?

A VAE is an autoencoder that is capable of learning the distribution in its latent space and thus generating new samples.

### 14. How does a VAE differ from a conventional autoencoder?

A traditional autoencoder learns a latent representation, while a VAE learns the latent distribution's parameters.

### 15. Why does a VAE learn a distribution instead of a single latent vector?

The distribution gives a continuous structure in the latent space where samples can be generated.

### 16. What is the reparameterization trick?

It represents the sampling process as  $z = \mu + \sigma \odot \epsilon$ .

### 17. Why should $\epsilon$ follow a standard normal distribution?

Using  $\epsilon \sim \mathcal{N}(0, I)$  separates the randomness and learned parameters  $\mu$  and  $\sigma$ .

### 18. What is the role of KL divergence in VAE loss function?

The KL divergence helps keep the learned latent distribution close to the selected prior distribution.

### 19. What is the prior distribution in VAE?

Prior distribution  $\mathcal{N}(0, I)$  is chosen.

### 20. Why is a two-dimensional latent space useful for visualization?

Two dimensions can be visualized directly; thus, it is easy to see clusters of overlapping latent digit representations.

### 21. What does latent space interpolation demonstrate?

Latent space interpolation shows whether transitioning between two latent points continuously also results in continuous transition of corresponding images.

22. **Why can a VAE generate new images?**

New images can be produced because new latent vectors can be sampled from the learned latent distribution and fed to the decoder.

23. **Why might some VAE-generated images be unclear?**

VAE generates samples randomly and therefore sometimes it samples latent points that lead to blurry digits.

24. **What is the difference between reconstruction and generation?**

Reconstruction tries to recreate an input image, while generation uses a sampled latent space to generate an entirely new image.

25. **Why should test images remain unseen during training?**

It allows for a good evaluation of generalization to unseen images.

## 14. Conclusion

In this experiment, we have been able to compare four different types of autoencoders for the purpose of image reconstruction and image generation. The FC-AE was capable of learning a compressed representation; however, due to its flattened structure, it was not efficient at capturing spatial features.

The CAE gave significantly superior results for reconstruction with its lower mean square error of 0.002682 and higher structural similarity index measure (SSIM) of 0.972842. The denoising CAE demonstrated that a convolutional autoencoder can reconstruct the main structure of the image even if the input image has been corrupted. The increase of Gaussian noise gradually increased the difficulty level of the task, evident from the higher reconstruction error and the lower SSIM values.

The VAE took a different approach and learned the probabilistic latent space. The two-dimensional latent space enabled visualization, and the random sampling and interpolation further indicated the generation capability of the model.

## 15. References

1. Y. LeCun, C. Cortes and C. Burges, "MNIST Handwritten Digit Database," <http://yann.lecun.com/exdb/mnist/>.
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3. D. P. Kingma and M. Welling, "Auto-Encoding Variational Bayes," arXiv:1312.6114.
4. F. Chollet, *Deep Learning with Python*, Manning Publications.
5. TensorFlow/Keras documentation, <https://www.tensorflow.org/>.